Outbreak Zero Grid Data

Copy the small grid first. Use the fixed 20×20 grid for testing; generate fresh random worlds for the investigation.

States and coordinates

0 empty (.) • 1 susceptible (H) • 2 infected (X) • 3 immune (I). Use world[row][column], starting at 0.

Small test grid

```
world = [
    [1, 1, 0, 1, 3],
    [0, 1, 1, 1, 0],
    [1, 1, 2, 1, 1],
    [3, 1, 1, 0, 1],
    [1, 0, 1, 1, 3]
]
```

Counts: 5 empty, 16 susceptible, 1 infected, 3 immune. Population 20; ever infected starts at 1.

At 100% transmission, 0% recovery and no hospitals: infect (1,2), (2,1), (2,3), (3,2). After one turn there are 5 infected; no diagonal or two-step spread.

Independent rows and snapshots

```
world = [[0 for column in range(20)] for row in range(20)]
```

```
old_world = [row[:] for row in world]
new_world = [row[:] for row in old_world]
```

Avoid [[0]*20]*20: it shares rows. Copying world[:] alone still shares inner rows. Read the old grid; write the new grid.

Fixed 20 by 20 test world

```
world = [
    [1, 1, 0, 3, 1, 1, 3, 0, 1, 0, 0, 0, 3, 1, 1, 1, 1, 1, 1, 1],
    [1, 1, 3, 1, 3, 1, 0, 3, 1, 0, 1, 0, 3, 1, 0, 1, 3, 0, 0, 1],
    [1, 1, 1, 0, 1, 0, 3, 1, 3, 1, 1, 1, 1, 0, 1, 0, 3, 1, 1, 0],
    [3, 1, 1, 1, 0, 0, 0, 3, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1],
    [1, 3, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0],
    [0, 1, 3, 3, 1, 1, 0, 1, 1, 1, 3, 1, 3, 3, 1, 1, 1, 1, 1, 1],
    [0, 1, 0, 0, 1, 3, 3, 0, 1, 1, 0, 1, 1, 0, 1, 1, 3, 3, 1, 0],
    [1, 1, 1, 0, 0, 1, 1, 1, 3, 1, 1, 3, 1, 1, 0, 3, 3, 1, 0, 0],
    [1, 3, 0, 3, 0, 0, 1, 0, 1, 0, 0, 1, 1, 3, 1, 0, 1, 1, 1, 1],
    [0, 1, 1, 3, 3, 3, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 3],
    [1, 1, 3, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 3, 1, 1, 0, 0, 1],
    [0, 0, 1, 1, 1, 0, 0, 0, 0, 3, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1],
    [0, 0, 1, 0, 0, 1, 0, 3, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 3, 0],
    [1, 1, 1, 1, 0, 1, 1, 1, 0, 3, 1, 0, 1, 1, 1, 3, 1, 1, 1, 1],
    [0, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1],
    [1, 1, 1, 1, 3, 0, 1, 3, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, 0, 3],
    [0, 1, 0, 3, 1, 1, 0, 2, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1],
    [1, 1, 1, 1, 3, 0, 1, 0, 1, 0, 3, 0, 1, 1, 0, 0, 0, 1, 3, 1],
    [1, 1, 3, 1, 0, 1, 3, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 3, 1, 1],
    [0, 1, 1, 1, 3, 0, 0, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 3, 3, 1]
]
```

Counts: 120 empty, 223 susceptible, 1 infected, 56 immune. Population 280; initial case (16,7). This is a fixed sample, not a random-trial generator.

Hospitals and settings

```
no_hospitals = []
central_hospitals = [[9, 9], [10, 10]]
edge_hospitals = [[0, 9], [19, 10]]

transmission_percent = 25
recovery_percent = 10
max_turns = 100
```

Hospitals do not replace people. Halve transmission into an orthogonally adjacent susceptible target once; effects do not stack. For experiments, use fresh worlds and the same recorded 100 seeds per condition.